Multi-View LP Bounds for Availability-LRCs

Delsarte Linear Programming Bounds in Coding Theory

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Getting Started

Background: Why Error-Correcting Codes?

- **Communication over Noisy Channels**:
  - Digital data (series of `0`s and `1`s) transmitted across networks can experience **bit flips** due to noise.
- **Core Idea**: Introduce controlled **redundancy** to detect and correct errors.
- **Simple Example (Repetition Code)**:
  - Message `0` $\rightarrow$ Send `000` | Message `1` $\rightarrow$ Send `111`
  - If receiver gets `010`:
    - Distance to `000` is 1; Distance to `111` is 2.
    - Guess the original message was `000`.

From the Example to the Language of Codes

- The two allowed messages are a subset of all length-three binary strings:

$$C = \{000, 111\} \subseteq \{0, 1\}^3$$

- The set C is a **code**; each member is a **codeword**.
- Its parameters are length $n = 3$, size $|C| = 2$, and minimum distance $d = 3$.

Connection: choosing a small subset of all strings creates room between codewords, and that room is what lets us correct noise.

Basic Definitions

- **Definition 1.1 (Code & Codeword)**
  Fix length $n \geq 1$ and an alphabet Σ. A **code** C is a subset $C \subseteq \Sigma^n$. Elements in C are called **codewords**.
  - **Example**: $C = \{\mathrm{BAG}, \mathrm{ACE}, \mathrm{FED}, \mathrm{EGG}\} \subseteq \{\mathrm{A}, ..., \mathrm{G}\}^3$.
- **Practical Setup**:
  - Usually set $\Sigma = \mathbb{F}_q$ (Finite Field with $q = p^m$) to utilize its **algebraic structure** (addition & multiplication).
  - If $q = 2, C \subseteq \mathbb{F}_2^n$ is a **binary code**.

Distances & Minimum Distance

Let $x, y \in \mathbb{F}_q^n$.

- **Definition 1.2 (Hamming Distance)**:

$$d_H(x, y) := |\{j \in \{1, ..., n\} : x_j \neq y_j\}|$$

- **Definition 1.3 (Hamming Weight)**:

$$\mathrm{wt}_H(x) := d_H(\mathbf{0}, x) = |\{j : x_j \neq 0\}|$$

- **Definition 1.4 (Minimum Distance)**:

$$d_{\min}(C) := \min_{x,y \in C, x \neq y} d_H(x, y)$$

  - Remark: $d_{\min}(C)$ measures the **fault tolerance**: a code with distance d can correct up to $\left\lfloor \frac{d-1}{2} \right\rfloor$ errors.

Classical Upper Bounds

Why Study Upper Bounds?

- Given code length n and minimum distance d, **how large can $|C|$ be?**
- **Objectives**:
  - Evaluate code efficiency (Information Rate).
  - Prove whether a discovered code is largest (when bound is sharp).
- **Classical Combinatorial Bounds**:
  - **Singleton Bound**: $|C| \leq q^{n-d+1}$
  - **Hamming (Sphere-Packing) Bound**: $|C| \leq \frac{q^n}{\sum_{j=0}^{\left\lfloor \frac{d-1}{2} \right\rfloor} \binom{n}{j} (q-1)^j}$
  - **Plotkin Bound** (for binary $2d > n$): $|C| \leq \frac{2d}{2d-n}$

What Do Classical Bounds Lack?

- **Limitation**:
  - Most classical bounds rely solely on **combinatorial counting** or **geometric volume** arguments.
  - They largely **ignore** the field arithmetic (additive and multiplicative group structures) of $\mathbb{F}_q^n$.
- **Solution**:
  - **Delsarte's Linear Programming Bound** bridges Harmonic Analysis (Fourier Transform on finite groups) and Linear Programming to yield vastly superior bounds.

Delsarte LP Bound

Difference Distribution

- Instead of looking at individual codewords, we study pair differences.
- **Definition 3.1 (Difference Distribution)**
  For $C \subseteq \mathbb{F}_q^n$, define $a_C : \mathbb{F}_q^n \to \mathbb{R}$ as:

$$a_C(x) := \frac{1}{|C|} \left|\{(y, z) \in C^2 : y - z = x\}\right|$$

  (Intuition: Average number of codeword pairs with difference x.)

- **Key Properties**:
  1. $a_C(\mathbf{0}) = 1$
  2. $a_C(x) \geq 0$ for all $x \in \mathbb{F}_q^n$
  3. $a_C(x) = 0$ for all $0 < \mathrm{wt}_H(x) < d_{\min(C)}$
  4. $\sum_{x \in \mathbb{F}_q^n} a_C(x) = |C|$

- **Definition 3.2 (Fourier Transform for Finite Abelian Groups)**
  For finite abelian group G, given $f : G \longrightarrow \mathbb{C}$, the Fourier transform $\hat{f} : \hat{G} \longrightarrow \mathbb{C}$ of f is defined by:

$$\hat{f}(\chi) = \sum_{x \in G} f(x)\overline{\chi(x)}$$

  where $\hat{G}$ is the set of all homomorphisms from G to $\mathbb{C}^{\times}$.

Connection to Fourier Analysis

- Notice that $a_{C(x)}$ can be rewritten as an **autocorrelation**:

$$a_C(x) = \frac{1}{|C|} \sum_{y \in \mathbb{F}_q^n} \mathbb{1}_C(y) \mathbb{1}_C(y - x)$$

- Applying Fourier Transform on Finite Abelian Group $\mathbb{F}_q^n$:

$$\widehat{a_C}(\chi) = \frac{1}{|C|} \mid \sum_{y \in \mathbb{F}_q^n} \mathbb{1}_C(y) \overline{\chi(y)} |^2 \geq 0$$

- **Fourier Non-negativity Constraint**:
  - The Fourier coefficients of a_C are **always non-negative**!

Singleton-level LP Bound

- We can formulate the code size bound as a **Linear Program**:

Maximize

$$\sum_{x \in \mathbb{F}_q^n} a(x)$$

Subject to

- $a(\mathbf{0}) = 1$
- $a(x) \geq 0, \quad \forall x \in \mathbb{F}_q^n$
- $a(x) = 0, \quad \forall x \text{ with } 0 < \mathrm{wt}_H(x) < d$
- $\hat{a}(\chi_u) \geq 0, \quad \forall u \in \mathbb{F}_q^n$

- **Computational Bottleneck**:
  - Number of variables = q^n (Exponential!). Impractical when $n \geq 10$.

Dimension Reduction via Krawtchouk Polynomials

- **Symmetry / Orbit Lumping**: Group elements with the same Hamming weight can be aggregated! $A_C(j) := \sum_{\mathrm{wt}_H(x)=j} a_C(x) = \frac{1}{|C|} \left|\{(y,z) \in C^2 : d_H(y,z) = j\}\right|$
- **Krawtchouk Polynomial**: Relates Fourier non-negativity directly to Hamming weights: $\sum_{j=0}^{n} A_C(j) K_k(j;n,q) \geq 0, \quad \forall k \in \{0,\ldots,n\}$ where

$$K_k(j;n,q) := \sum_{\ell=0}^{k} (-1)^\ell (q-1)^{k-\ell} \binom{j}{\ell} \binom{n-j}{k-\ell}$$

- **Resulting Delsarte LP Bound**:
  - Variables reduced from q^n to just $\boldsymbol{n+1}$!
  - Extremely efficient to solve numerically via LP solvers.

The Delsarte LP: One Variable Per Distance

For a code with minimum distance at least d, maximize the total distance-distribution mass:

Maximize

$$\sum_{j=0}^{n} A(j)$$

Subject to

- $A(0) = 1$
- $A(j) \geq 0 \quad (0 \leq j \leq n)$
- $A(j) = 0 \quad (1 \leq j < d)$
- $\sum_{j=0}^{n} A(j) K_k(j; n, q) \geq 0 \quad (0 \leq k \leq n)$

Each real code gives a feasible point, so the LP maximum is an upper bound on $|C|$.

Delsarte LP Bound vs. Other Bounds

- For binary codes:

(n, d)	Delsarte LP	Hamming	Plotkin	Singleton	Exact Value
$(5, 3)$	4	5	6	8	4
$(7, 4)$	8	16	8	16	8
$(9, 4)$	25	51	—	64	20
$(11, 4)$	85	170	—	256	72
$(13, 4)$	292	585	—	1024	256

Locally Recoverable Codes (LRCs)

Why Locally Recoverable Codes?

- **Distributed Storage Problem**:
  - Modern cloud storage (e.g., Google File System, AWS S3) stores petabytes of data.
  - Standard codes require reading $O(k)$ nodes to repair **a single broken node** $\rightarrow$ High I/O overhead.
- **Local Recovery Goal**:
  - Repair any corrupted symbol by reading only a **small set** $r \ll k$ of other symbols (Local View).
- **Availability**:
  - Multiple disjoint repair sets for parallel recovery and high request traffic handling.

Local-distance Availability

A code has all-symbol local-distance (r, δ, a)-availability if, for every coordinate i, we can choose a views

$$T_{i,h} = \{i\} \cup S_{i,h}, \quad h = 1, ..., a$$

such that

- $|T_{i,h}| \leq r + \delta - 1$,
- the projected code $C|_{T_{i,h}}$ has minimum distance at least δ,
- the helper sets $S_{i,1}, ..., S_{i,a}$ are pairwise disjoint.

Since $\delta \geq 2$, each view can recover an erased coordinate from the other symbols in that view. This local-distance condition is stronger than functional recovery alone. [1]

Partition around Each Coordinate

For two available views, fix a coordinate i and split all coordinates into four disjoint blocks:

$$S_{i,1} \quad S_{i,2} \quad \{i\} \quad O_i$$

$$[n] = S_{i,1} \cup S_{i,2} \cup \{i\} \cup O_i$$

For an ordered pair of codewords (x, y), record four distances:

- $j = d_{S_{i,1}}(x, y)$ and $k = d_{S_{i,2}}(x, y)$,
- $b = 1_{x_i \neq y_i}$ and $c = d_{O_i}(x, y)$.

These satisfy the exact identity $d_H(x, y) = j + k + b + c$.

The Multi-view Distribution

Let $M = |C|$, $t_h = |T_{i,h}|$, and $u = t_1 + t_2 - 1$. Average over ordered pairs and recovered coordinates:

$$Z_C(t_1, t_2, j, k, c, b) = \frac{1}{nM} \left| \left\{ (x, y, i) : |T_{i,h}| = t_h, d_{S_{i,1}}(x, y) = j, d_{S_{i,2}}(x, y) = k, d_{O_i}(x, y) = c, 1_{x_i \neq y_i} = b \right\} \right|$$

- The index b separates pairs that agree or differ at the recovered coordinate.
- The outside block has length $n - u$, so $0 \leq c \leq n - u$.
- The global distance is already encoded: $\ell = j + k + c + b$.

Four Constraints

Constraint family	What it enforces
Centering	The local view counts must agree with the global distance distribution.
Distance support	Pairs cannot be too close globally or inside a local view.
Product-Krawtchouk	Every four-block frequency has nonnegative Fourier energy.
Collisions	Small local projections force many codeword pairs to agree there.

The Actual Multi-view LP: Variables and Objective

Fix q, n, d, r, δ and $a = 2$. Let $u = t_1 + t_2 - 1$.

Variables

- $Z(t_1, t_2, j, k, c, b) \geq 0$ on admissible indices:
- $\delta \leq t_1, t_2 \leq \min(n, r + \delta - 1), \quad t_1 + t_2 - 1 \leq n$
- $0 \leq j < t_1, \quad 0 \leq k < t_2, \quad 0 \leq c \leq n - u, \quad b \in \{0, 1\}$

Global distribution

- $A(\ell) := \sum_{t_1, t_2} \sum_{j+k+c+b=\ell} Z(t_1, t_2, j, k, c, b)$

Goal: Maximize $\sum_{t_1, t_2, j, k, c, b} Z(t_1, t_2, j, k, c, b) = \sum_{\ell=0}^{n} A(\ell)$

Each real code induces one feasible assignment of these variables.

Subject to Global Counting

For a pair at distance ℓ, exactly ℓ of the n possible recovered coordinates differ. The centered identities split the global mass according to b:

$$\sum_{t_1,t_2} \sum_{j+k+c=\ell} Z(t_1,t_2,j,k,c,0) = \frac{n-\ell}{n} A(\ell)$$
$$\sum_{t_1,t_2} \sum_{j+k+c+1=\ell} Z(t_1,t_2,j,k,c,1) = \frac{\ell}{n} A(\ell)$$

The global distance distribution also satisfies:

$$A(0) = 1, \quad A(\ell) \geq 0, \quad A(\ell) = 0 \quad \text{for} \quad 1 \leq \ell < \max(d, \delta)$$

Subject to Distance Support

Short distances are forbidden by the global distance and the local-distance requirement. Write $d^\star = \max(d, \delta)$.

Local views

$$Z(t_1, t_2, j, k, c, b) = 0 \quad \text{if} \quad 1 \le j + b < \delta$$

$$Z(t_1, t_2, j, k, c, b) = 0 \quad \text{if} \quad 1 \le k + b < \delta$$

Global code distance

$$Z(t_1, t_2, j, k, c, b) = 0 \quad \text{if} \quad 1 \le j + k + c + b < d^\star$$

The first two conditions protect each repair view; the last excludes globally too-close codeword pairs.

Subject to Product-Krawtchouk Positivity

Every frequency tuple gives a nonnegative transform of the four-block distribution:

$$P(j,k,c,b) := K_{p_1}(j;t_1-1,q)K_{p_2}(k;t_2-1,q)$$
$$\times K_{\eta}(c;n-u,q)K_{e}(b;1,q)$$
$$\Phi_C(t_1,t_2;p_1,p_2,\eta,e) := \sum_{j,k,c,b} Z_C(t_1,t_2;j,k,c,b)P(j,k,c,b) \geq 0$$

$$0 \leq p_1 < t_1, \quad 0 \leq p_2 < t_2, \quad 0 \leq \eta \leq n-u, \quad e \in \{0,1\}$$

This is imposed for every admissible (t_1, t_2) and every frequency tuple in these ranges.

Why Does Locality Force Collisions?

Project the M codewords onto a length-L set whose projected code has distance at least δ. Singleton says there are at most $q^{L-\delta+1}$ distinct projections.

If f_z codewords have projection z, then Cauchy-Schwarz gives:

$$q^{L-\delta+1} \sum_z f_z^2 \geq M^2$$

This is the normalized number of ordered pairs that **collide** on the projection.

- Apply it to the first local view.
- Apply it to the second local view.
- Apply it to the union of both helper sets (with the recovered coordinate removed).

Subject to Local Collision Inequalities

Let $m(t_1, t_2) = \sum_{j,k,c,b} Z(t_1, t_2, j, k, c, b)$ be the total mass for a fixed pair of view lengths.

First local view

$$q^{t_1-\delta+1} \sum_{k,c} Z(t_1, t_2, 0, k, c, 0) \geq m(t_1, t_2)$$

Second local view

$$q^{t_2-\delta+1} \sum_{j,c} Z(t_1, t_2, j, 0, c, 0) \geq m(t_1, t_2)$$

Punctured union of helper sets

$$q^{t_1+t_2-\delta-1} \sum_{c} Z(t_1, t_2, 0, 0, c, 0) \geq m(t_1, t_2)$$

The first two rows use each full repair view; the last uses both helper sets with the recovered coordinate removed.

(q, n, d, r, δ)	Multiview	Delsarte	Verified Code Size
$(2, 9, 3, 1, 2)$	8	42	8
$(2, 14, 4, 2, 2)$	90.35	512	64
$(3, 8, 3, 2, 3)$	9	340	9
$(3, 9, 5, 2, 2)$	27	90	27
$(4, 8, 4, 2, 2)$	64	614	64

Numerical Results & Key Insights

- **Main Contributions**:
  1. Extended classical Delsarte LP to high-dimensional local distance distributions.
  2. Derived new upper bounds for $(r, \delta, a = 2)$-availability LRCs.
  3. Numerical computations show our Multi-View LP bound strictly improves upon existing single-view bounds for intermediate blocklengths.